

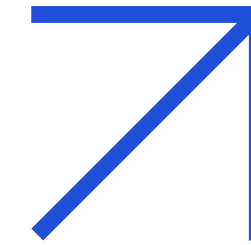
Thynk Unlimited

NETFLIX MOVIES & TV SHOWS DASHBOARD

Data Visualization Final Project | Manish
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| University of Europe for Applied Sciences
(UE) | July 30, 2026



PROJECT OVERVIEW



Purpose & Goals

This project builds an interactive Netflix Movies & TV Shows Dashboard to explore content trends and deliver meaningful insights through dynamic visualizations. The Netflix dataset was chosen for its diversity, popularity, and rich metadata. The goal is to create a user-friendly experience that makes data exploration intuitive and engaging.

DATASET OVERVIEW



Netflix Movies & TV Shows Dataset

Source: Kaggle • ~8,800+ records of Movies and TV Shows

Key columns: Title, Type, Director, Country, Genre, Rating, Duration, Release Year, Date Added.

Rich, diverse, and temporal data makes this dataset ideal for interactive visualization and analysis.

DATA CLEANING & PREPROCESSING



1

Removed Duplicate Records

2

Filled Missing Values

3

Converted Date Columns

4

Created New Features

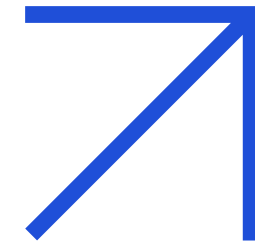
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Prepared Dataset for Visualization

Cleaning Workflow

A structured pipeline that transforms raw Netflix data into a clean, analysis-ready dataset for the dashboard.

TECHNOLOGIES USED

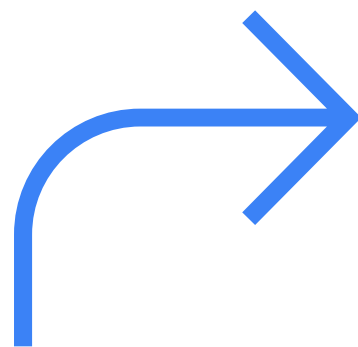


Python:

Core programming language for data processing

Pandas:

Data manipulation and cleaning



Plotly:

Interactive charting library

Streamlit:

Dashboard development and deployment



DASHBOARD FEATURES



1

Interactive KPI Cards

2

Sidebar Filters

3

Interactive Plotly Charts

4

Dataset Preview

5

Responsive Layout & Filtering

Core Functionalities

A user-friendly dashboard combining dynamic filters, real-time updates, and interactive visualizations for a seamless Netflix data exploration experience.

DASHBOARD VISUALIZATIONS

Movies vs TV Shows:

Compare content types across the library

Content Growth:

Track Netflix additions over years

Top Countries:

See which nations produce the most titles

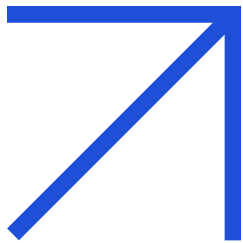
Top Genres:

Discover the most popular categories



KEY INSIGHTS

FROM THE DASHBOARD

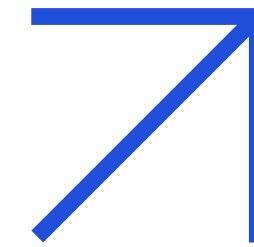


- 1 Movies significantly outnumber TV Shows in the Netflix library
- 2 Rapid content growth observed after 2015
- 3 United States contributes the highest number of titles
- 4 International Movies and TV-MA ratings are most frequent
- 5 Monthly content additions show clear seasonal trends

Major Findings

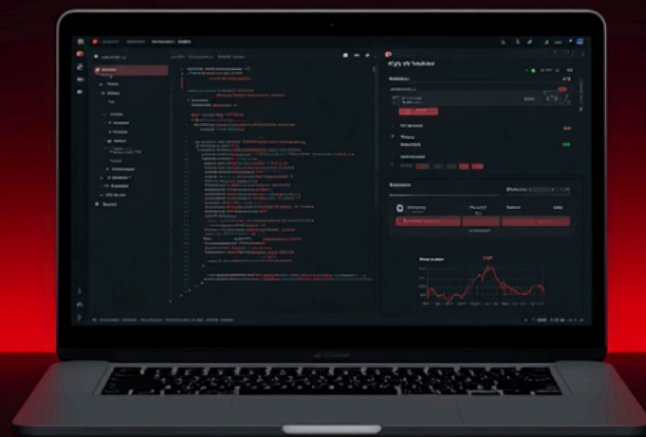
The Netflix Movies & TV Shows Dashboard reveals clear patterns in content distribution, growth trends, and audience targeting—turning raw data into actionable insights.

DEPLOYMENT PROCESS



How the Dashboard Went Live

The project was developed in Jupyter Notebook, then built into an interactive dashboard with Streamlit. The source code is hosted on GitHub and deployed online using Streamlit Community Cloud, making it accessible anywhere.



CHALLENGES & SOLUTIONS



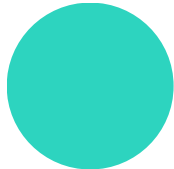
Handling missing values →
Implemented robust cleaning
methods



Data preprocessing complexities →
Developed automated pipelines



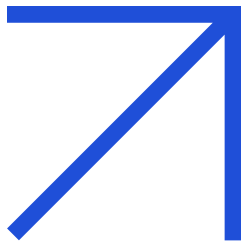
Plotly visualization issues →
Adjusted chart configurations



Streamlit deployment issues →
Debugged and optimized scripts

FUTURE IMPROVEMENTS

ENHANCING THE DASHBOARD



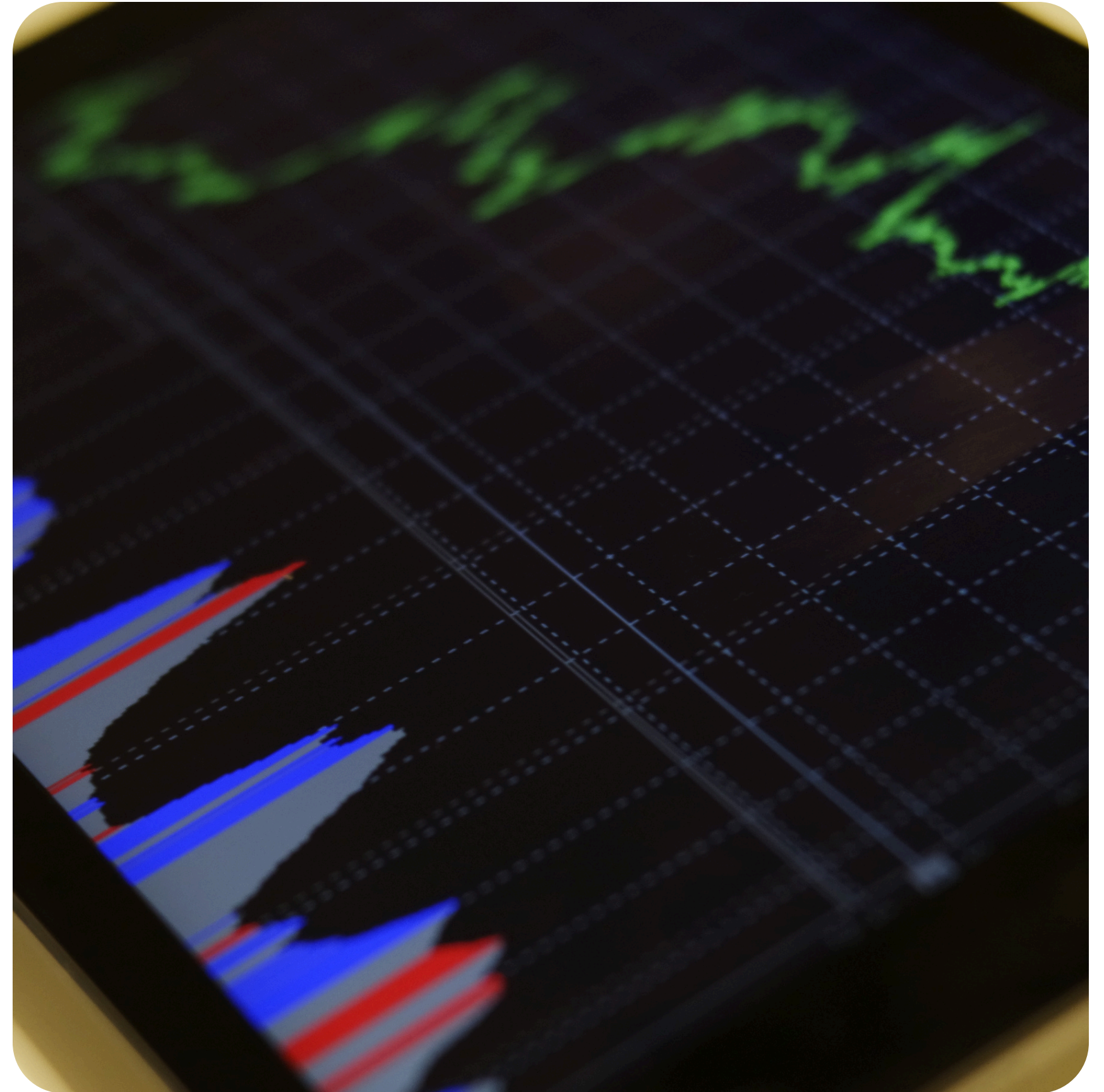
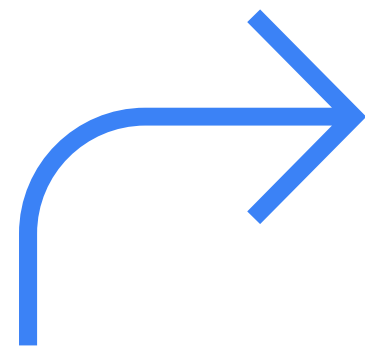
- 1 Search Functionality for Titles
- 2 Recommendation System for Personalized Suggestions
- 3 Machine Learning for Predictive Analytics
- 4 Live Netflix API Integration for Real-Time Data
- 5 Additional Interactive Visualizations for Deeper Insights

Planned Enhancements

Next steps to make the dashboard more powerful, personalized, and connected to real-time data sources.

Netflix Dashboard | Data
Visualization Final Project

THANK YOU



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